ResumeIQ

- **Project Name:** ResumeIQ
- **Tagline:** Analyze. Improve. Get hired.
- **Team Name:** Van Guard
- **Event:** DevCraft: Simon’s Command Line
- **Date**: 06-04-2026
- **Domain:** AI

Problem Statement

- Modern hiring is dominated by **Applicant Tracking Systems (ATS)**.
- 75%+ resumes are rejected before reaching recruiters.
- Candidates lack knowledge of ATS formatting & keyword optimization.
- Generic resumes fail to match job descriptions effectively.
- Students & freshers especially struggle with resume personalization.

Target Audience

- College Students & Fresh Graduates
- Job Seekers (Entry & Mid-Level)
- Career Switchers
- Professionals applying to MNCs
- Placement Cell Candidates

Existing Solutions

- Basic resume templates (Canva, MS Word)
- LinkedIn resume builder
- Static resume review platforms
- Generic AI text generators

Gaps & Opportunities

- No real ATS compatibility scoring
- Limited keyword-job matching
- No actionable improvement suggestions
- Not personalized to specific job descriptions
- Poor formatting feedback

Opportunity: Build an intelligent, job-specific, AI-powered resume optimizer.

Your Solution

ResumeIQ is an **AI-driven ATS Resume Analyzer** that:

- Analyzes resumes using NLP techniques
- Compares resume with job description
- Calculates ATS compatibility score
- Suggests missing keywords
- Provides personalized improvement insights

Key Features

- ATS Score Generator
- Job Description Matching
- Keyword Gap Analysis
- Resume Improvement Suggestions
- PDF/Text Upload Support
- Clean & Simple UI
- Real-time Feedback

System Architecture

Frontend:
User uploads resume & job description

Backend:
Resume Parsing
Text Preprocessing
NLP Keyword Extraction
Similarity Matching Algorithm
ATS Scoring Engine

Output:
ATS Score
Keyword Match Percentage
Suggestions for Improvement

Tech Stack

- **Frontend:**

HTML
CSS (animation + transitions)
JavaScript
Chart.js (for analytics visualization)
AOS (for UI animations)

- **Backend:**

Python
FastAPI

- **AI/NLP:**

SpaCy (keyword + entity extraction)
OpenAI
Sentence Transformers (semantic similarity)

- **PDF Generation:**

WeasyPrint

Workflow

- User uploads Resume (PDF/Text)
- User pastes Job Description
- System extracts text
- NLP preprocessing (stopwords removal, tokenization)
- Keyword extraction
- Similarity score calculation
- ATS score generation
- Display suggestions

Impact

- Increases chances of resume shortlisting
- Helps students crack campus placements and entry level jobs
- Reduces resume rejection rate
- Saves time in manual resume editing
- Makes job applications data-driven

Future Scope

- AI-powered resume rewriting
- Industry-specific resume templates
- LinkedIn profile optimization
- Resume analytics dashboard
- Integration with job portals
- Machine Learning-based scoring model

Team Info

Magizharasi : Testing & Data Engineer

Richie : Backend & API Developer

Shwetha : Frontend Developer

Nanditha : AI/ML Engineer